

Sustainable design

In light of environmental concerns, there are increasing calls for architecture to be sustainable or green. Over the lifespan of a large building, the processes of construction, maintenance and demolition use a lot of energy and resources, alongside the energy used for heating, cooling and lighting in normal operation. But where buildings are concerned, younger doesn't always mean greener: a 2012 report on New York City's buildings found that the Empire State Building (1931) was more efficient than the supposedly 'green' 7 World Trade Center (2008). Much modern architecture is energy-inefficient on account of its expansive glazing and thin walls, and a frequent prioritization of style over resilient or sustainable design. For architecture to retain its credibility, it must conserve the Earth's finite resources, reduce energy consumption, manage waste more effectively and create buildings that work with the environment. Sydney's One Central Park (opposite, 2010–13) for example, incorporates natural lighting technology, 'hanging gardens', and other sustainable technology such as rainwater retention and recycling systems.

